YIGON KIM

Ph.D. Student in Astronomy, Seoul National University
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Research Interests

Galactic dynamics and mass reconstruction; dark matter and gravitational inference; astrometric time-domain analysis; cosmological structure formation; statistical inference from large observational and simulation data sets.

Education

Seoul National University **Mar 2024–Feb 2027 (expected)**
Ph.D. in Astronomy *Seoul, Republic of Korea*

- Dissertation: *Mapping the Three-Dimensional Mass Distribution of the Milky Way in the Extended Solar Neighborhood with Gaia.*

Kyungpook National University **Mar 2019–Feb 2021**
M.Sc. in Astronomy *Daegu, Republic of Korea*

Kyungpook National University **Mar 2015–Feb 2019**
B.Sc. in Astronomy and Atmospheric Sciences; Minor in Physics *Daegu, Republic of Korea*

Research Experience

Seoul National University **Mar 2024–present**
Ph.D. Researcher, Extragalactic Astronomy and Cosmology Team *Seoul, Republic of Korea*

- Develop methods to infer three-dimensional matter and gravitational fields from stellar astrometry by combining Gaia observations, cosmological simulations, and statistical or machine-learning models.
- Built end-to-end pipelines for volumetric data construction, GPU training, uncertainty estimation, simulation–observation comparison, and validation of inferred dark-matter density fields.
- Conduct ongoing work on astrometric acceleration, gravitational-potential reconstruction, and density recovery using numerical differentiation and Gaia-like time-domain data.

Kyungpook National University **Sep 2022–Dec 2023**
Researcher, SPHEREx Basic Research Laboratory (NRF-funded) *Daegu, Republic of Korea*

- Assessed the potential impact of galaxy–galaxy blending on SPHEREx observations by constructing controlled samples, comparing simulations with observations, and conducting spatial-statistical analyses in collaboration with the research team.

Korea Astronomy and Space Science Institute **Sep–Oct 2020**
Visiting Researcher *Daejeon, Republic of Korea*

- Studied secondary halo bias and the relation between galaxy motion and dark-matter halo spin using cosmological simulations and large-scale galaxy catalogs.

Publications

1. **Y. Kim**, H. S. Hwang, S. E. Hong, D. Jeong, M. Yang and J.-h. Kim. "Three-dimensional dark matter mapping in the solar neighborhood with Gaia Data Release 3 and deep learning." Submitted to *Monthly Notices of the Royal Astronomical Society* (2026).
2. **Y. Kim**, A. D. Montero-Dorta, R. Smith, and J.-H. Shinn. "Towards an Observational Detection of Halo Spin Bias using Spin-Orbit Coherence." *Astronomy & Astrophysics*, **700**, A149 (2025).
3. **Y. Kim**, R. Smith, and J. Shin. "Unexpected Dancing Partners: Tracing the Coherence between the Spin and Motion of Dark Matter Halos." *The Astrophysical Journal*, **935**, 71 (2022).
4. **Y. Kim**, C.-G. Park, H. Noh, and J.-c. Hwang. "CMASS Galaxy Sample and the Ontological Status of the Cosmological Principle." *Astronomy & Astrophysics*, **660**, A139 (2022).

Selected Presentations

- "Machine Learning Inference of the Local 3D Dark Matter Map from Gaia DR3 Astrometry." Korean Astronomical Society (KAS) Fall Meeting (Poster, 2026).
- "Coherent Motion of Galaxies as the Proxy of Secondary Halo Bias." KAS Fall Meeting (Oral, 2023).
- "CMASS Galaxy Sample and the Ontological Status of the Cosmological Principle." KAS Fall Meeting (Master's degree presentation, 2020).
- "A Visual Investigation of Cosmic Inhomogeneity Using the CMASS Galaxy Sample." KAS Spring Meeting (Oral, 2018).

Grants and Awards

2026–present	Support for Next Generation Academics, Seoul National University
2024–2025	BK21 Fellowship and two Seoul National University merit scholarships
2015–2018	Merit scholarships, Kyungpook National University

Academic Service and Research Infrastructure

SNU Extragalactic Astronomy and Cosmology Team **Mar 2025–present**
GPU and CPU Server Manager *Seoul, Republic of Korea*

- Administer Linux-based shared computing infrastructure, including SLURM workloads, GPU resources, storage, user environments, and operational troubleshooting.

SNU Astronomy Program **Mar 2025–Feb 2026**
Graduate Student Representative *Seoul, Republic of Korea*

Korean Astronomical Society **2017–present**
Member *Republic of Korea*

Technical Skills

Scientific computing	Python, PyTorch, FORTRAN, MPI, Linux/shell, SLURM, GPU computing
Data analysis	Large-scale astronomical catalogs, cosmological simulations, volumetric and phase-space data, astrometry, numerical method, statistical inference
Methods	Deep learning, simulation–observation comparison, uncertainty estimation, spatial statistics, scientific visualization